



Two cases of fatal insulin homicide resolved using combined immunohistochemistry and immunofluorescence techniques

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Abstract

Diabetes is a prevalent disease in modern society. Since the development and widespread use of synthetic insulin, its therapeutic application has become globally recognized for managing diabetes. However, excessive administration of insulin may lead to fatal outcomes. The rapid pharmacokinetics of insulin in the human body, combined with the lack of distinct pathological changes, makes the forensic identification of fatal insulin overdose challenging. In this study, we present two cases of fatal insulin overdose, one with a postmortem interval of up to 28 days. Postmortem specimens were collected and analyzed. Immunohistochemical and immunofluorescence assays revealed the consistent distribution of insulin and protamine at the injection sites in both cases. Insulin and protamine particles were observed in the epidermis, epidermal cells, adipocyte membranes, inter-adipocyte spaces, inflammatory cells, and glandular structures. Toxicological analyses confirmed the presence of insulin in the injected skin, subcutaneous fat, blood, urine, vitreous humor, and bile. This study provides valuable insights and guidelines for the forensic identification of insulin overdose cases.

Keywords Insulin overdose · Immunohistochemistry · Immunofluorescence · Protamine

Introduction

In healthy, non-obese individuals, fasting plasma insulin concentrations range between 0.2 and 0.5 ng/mL and can rise to as high as 3 ng/mL following glucose stimulation [1]. Insulin plays a pivotal role in regulating human metabolism at physiological concentrations [1–3]. However, excessive insulin administration can result in severe hypoglycemia, a potentially life-threatening condition. The International Hypoglycemia Study Group defines clinically significant hypoglycemia as a glucose concentration of less than 3.0 mmol/L (< 54 mg/dL), a threshold associated with cognitive dysfunction, arrhythmias, and in extreme cases, death [4].

The earliest documented case of attempted suicide using exogenous insulin was reported in 1927 [5]. The first recorded instance of insulin-induced homicide appeared in the literature in 1958 [6]. It was not until the 1950s, with the introduction of radioimmunoassay, that the ability to detect insulin levels in forensic contexts significantly improved, enabling the accurate identification of insulin overdose as a cause of death [7, 8]. Since then, numerous cases of fatal insulin overdose have been reported [9–20]. To date, the primary methods for identifying fatal insulin overdoses rely on traditional histopathological techniques,

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Table 1 Basic information of the two cases

	Case 1	Case 2
Age/sex	47/Male	49/Female
Perpetrator	Mother	Lover
Injection method	Subcutaneous and Intravenous injection	Subcutaneous injection
Positions of injection sites	Right abdomen×4, left elbow×1	Abdomen×4
Total injection volume	>900U	>300U(3mL)
Type of insulin	Insulin (with protamine)	Insulin Aspart
Survival time	21 h~34 h	3 h~8 h
Postmortem interval before autopsy	The first autopsy:6 days The second autopsy:28 days	<2 days
Decedent histories	Schizophrenia (No diabetes)	Not Found (No diabetes)

such as hematoxylin and eosin (HE) staining, in combination with radioimmunoassay [19, 21]. In this study, we report two homicide cases where insulin overdose was the cause of death. We performed immunohistochemistry and dual immunofluorescence labeling on skin samples from the insulin injection sites. Additionally, we quantitatively measured insulin levels in cadaveric tissues using radioimmunoassay, providing robust evidence that insulin overdose was the definitive cause of death.

Case reports

Case 1

A psychiatric patient had a long-standing history of conflict with his mother. One evening, the mother dissolved 30 chlorpromazine tablets in water and administered the

solution to the patient. After the patient fell asleep, she injected large doses of insulin into his abdominal skin on multiple occasions. The patient awoke at 7 a.m. the following day, ate breakfast, and then returned to sleep at 6 p.m. On the third morning after the insulin injections, the mother discovered the patient deceased.

Case 2

The perpetrator and the victim were engaged in an extra-marital affair. Following a heated argument, the perpetrator coerced the victim into consuming an unknown quantity of sedatives. After the victim had fallen asleep, the perpetrator administered several large doses of insulin by injecting into the victim's abdominal skin multiple times. Three hours later, upon realizing the victim was still alive, the perpetrator injected additional insulin, though the exact amount was unspecified. The following morning, the perpetrator found the victim dead.

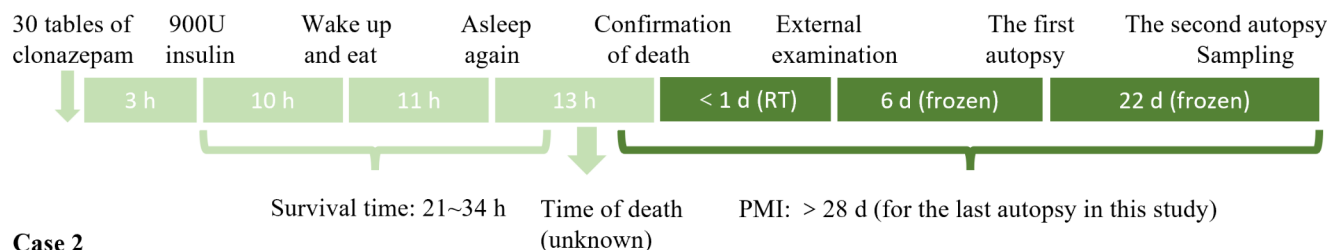
The basic information about the two cases is listed in Table 1, and the timelines of both cases are shown in Fig. 1.

Autopsy findings and histopathological examinations

Skins from insulin injection sites

In Case 1, five injection sites were identified. Wounds No. 1 and No. 2, located on the abdomen, were discovered during the initial autopsy six days postmortem (Fig. 2a). Wounds No. 3 and No. 4 on the abdomen, and Wound No. 5 on the left elbow, were identified during the second autopsy 28 days after death (Fig. 2b). In Case 2, four injection sites were

Case 1



Case 2

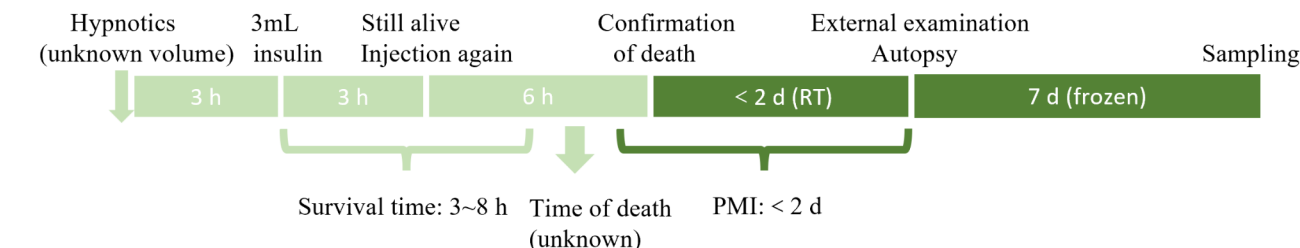


Fig. 1 Timelines of the two cases

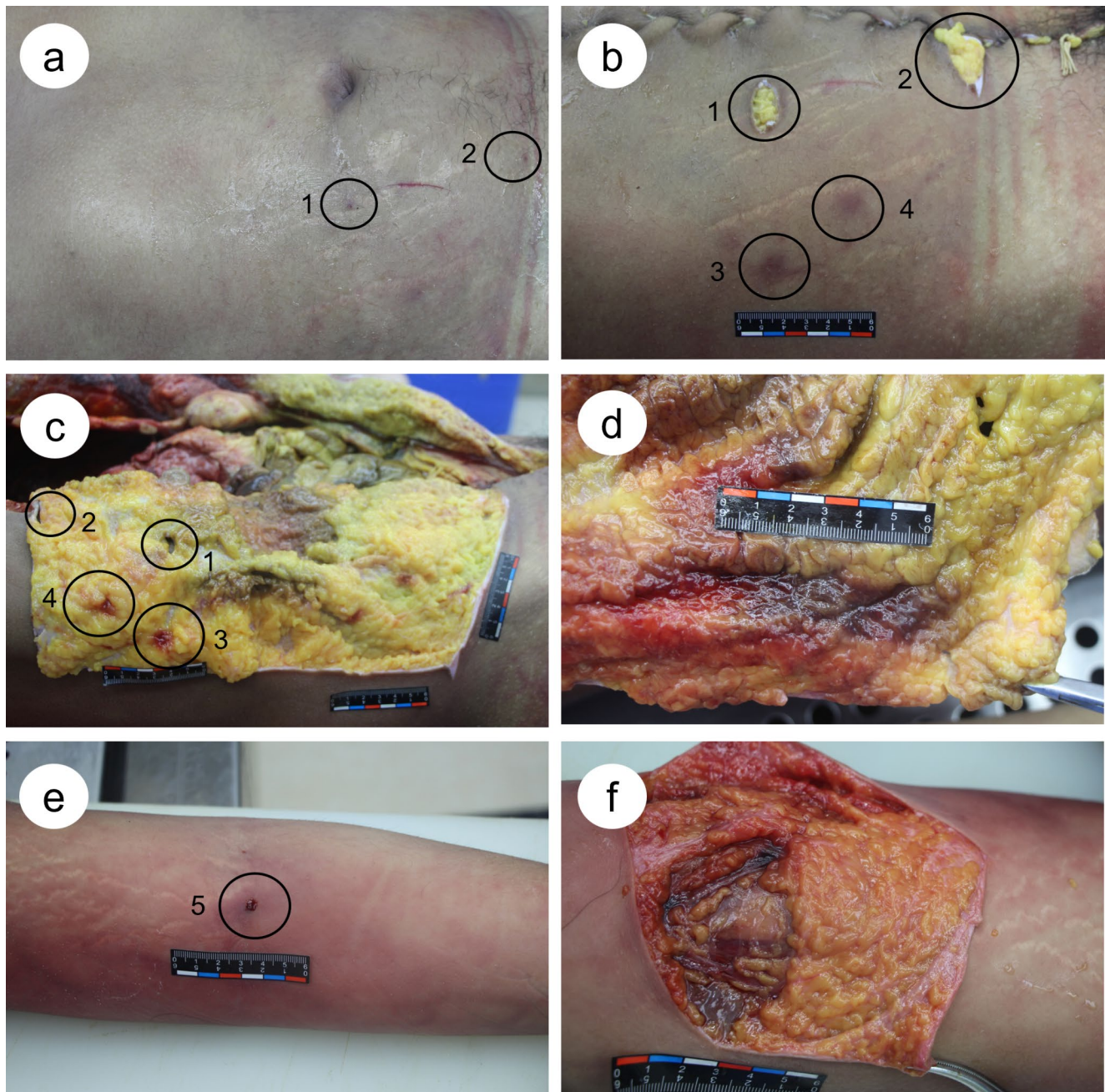


Fig. 2 Injection marks in Case 1. **a**: Wound No. 1 and No. 2 in the abdomen revealed in the first autopsy; **b**: Wound No. 3 and No. 4 in the abdomen revealed in the second autopsy (tissue samples had been collected from Wound No. 1 and No. 2); **c**, **d**: Patchy hemorrhage in

found in the abdominal area (Fig. 3a-c). Upon dissection of the skin and subcutaneous tissues at these sites, marked local edema and multiple subcutaneous hemorrhages were noted (Figs. 2c-f and 3e-f). Microscopic analysis revealed significant hemorrhage in the subcutaneous adipose tissue. In Case 1, lymphocytes were the predominant inflammatory cells, whereas in Case 2, a substantial neutrophil infiltration was observed (Fig. 4b). Additionally, pink fluid accumulated

the superficial and deep subcutaneous adipose tissue at the abdominal injection site; **e**, **f**: Wound No. 5 at the left elbow with subcutaneous hemorrhage and edema

in the interstices of some adipocytes, and certain collagen fibers appeared swollen and stained red (Fig. 4a).

Brain

Neurons exhibited widespread degeneration and necrosis, with pyknotic nuclei and red-stained cytoplasm, predominantly affecting the cerebral cortex (Fig. 4c). Astrocytes

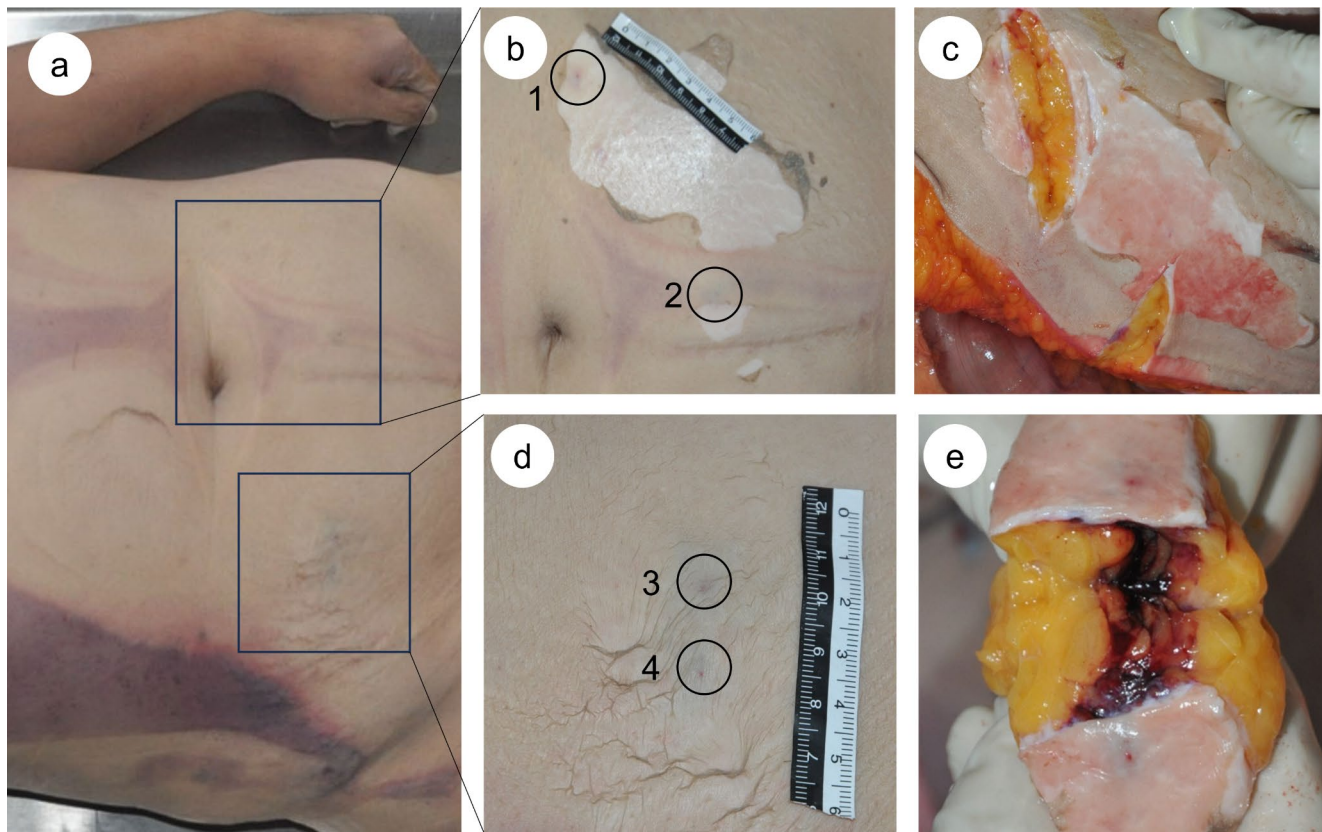


Fig. 3 Injection marks in Case 2. **a**: 4 abdominal injection sites; **b**, **c**: Wound No.1 in the left abdomen and No.2 in the lower abdomen without obvious hemorrhage; **d**, **e**: Wound No.3 and No.4 in the right abdomen with subcutaneous hemorrhage

showed reactive hypertrophy and proliferation throughout the brain accompanied by edema. An increased number of inflammatory cells were present in the subarachnoid space vessels.

Heart

Inflammatory cells infiltration in the myocardial interstitial blood vessels was increased, along with interstitial edema. In Case 2, a mild lymphocytic infiltration was observed in the myocardium.

Lungs

Several alveoli were filled with pink, proteinaceous edema fluid, and some alveoli exhibited dilation, consistent with emphysema. Scattered macrophages infiltration was present, and the alveolar blood vessels were markedly congested.

Liver

Hepatocytes exhibited edema and steatosis (Fig. 4d), with scattered lymphocytic infiltrates in the hepatic portal area. The hepatic sinusoids were congested.

Kidneys

Necrosis of epithelial cell was observed in both the proximal and distal renal tubules, with partial detachment into the tubules lumens.

Furthermore, congestion, edema, and substantial inflammatory cell infiltration were observed across multiple organs. The histopathological findings of the major organs are summarized in Table 2.

Insulin analysis

Skins samples from the insulin injection sites in both cases were collected as experimental specimens, while skins samples distant from the injection sites served as controls. Immunohistochemical (IHC) assays for insulin/protamine and double immunofluorescence (IF) staining were performed. Insulin and C-peptide concentrations were measured in the skin at the injection sites, corresponding subcutaneous adipose tissues at various depths, and bodily fluids including blood, urine, vitreous humor, and bile. The detection method employed was radioimmunoassay (RIA).

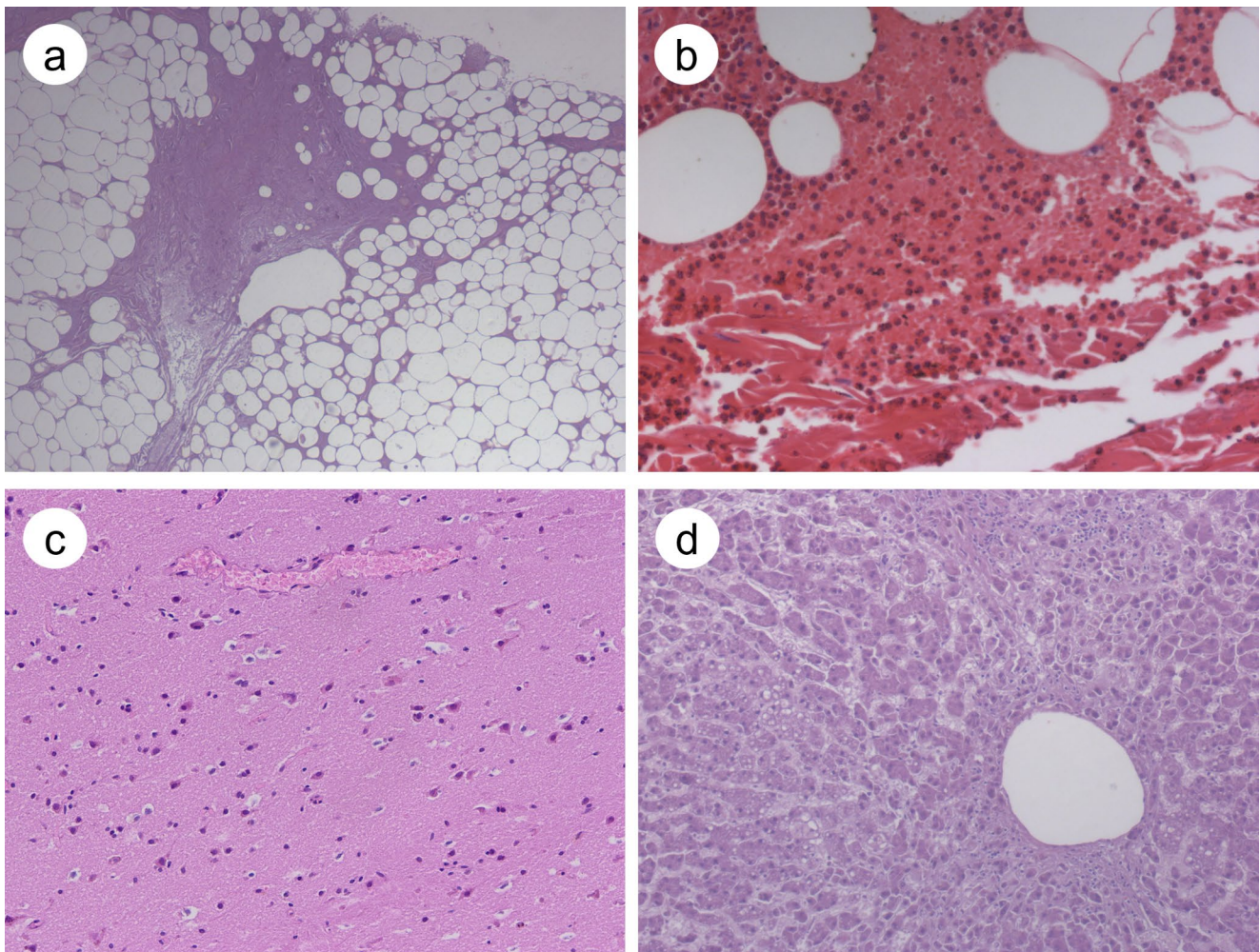


Fig. 4 Histopathologic changes, H&E. **a:** Pink fluid accumulation in the subcutaneous adipose interstices at the injection site, 1:40; **b:** Significant hemorrhage with neutrophil infiltration in the skin at the

injection site, 1:200; **c:** Neuronal degeneration/necrosis in the cerebral cortex, 1:100; **d:** Hepatocellular steatosis, 1:100

Immunohistochemistry

Positive IHC staining for insulin and protamine were observed in the skin tissue from the injection sites in both cases. Insulin and protamine were primarily localized in adipocyte membranes (Fig. 5a), fat spaces (Fig. 5b), epidermal layers (Fig. 5c, d), sebaceous glands (Fig. 5e, f), inflammatory cells (Fig. 5g), and sweat glands (Fig. 5h). Skin samples distant from the injection sites and those not exposed to insulin showed negative staining results.

Double immunofluorescence staining

Nearly identical distributions of insulin and protamine were observed in the sweat glands (Fig. 6a-c) and fat spaces (Fig. 6d-g).

Toxicological findings

Significant insulin levels were detected in biological samples from both cases. In Case 1, the highest insulin concentration was found in the subcutaneous adipose tissue of the right abdomen, while in Case 2, elevated insulin levels were detected in the injection site skin and bile. Psychotropic drugs were also detected in the blood of both deceased individuals. Detailed sample information and the results are provided in Table 3.

Based on the autopsy, histopathological examination, and toxicological analysis, insulin overdose was confirmed as the cause of death in both case.

Table 2 Histopathologic changes of 2 cases

Histopathologic changes		Case 1	Case 2
Skin at the injection sites	Hemorrhage	++	+++
	Inflammation	+	+++
	infiltration	Lymphocyte	Neutrophil
	Pink fluid in the fat space	++	+++
Brain	Necrotic neurons	++	++
	Astrocyte proliferation	++	++
Heart	Inflammation infiltration	-	+ Lymphocyte
Lungs	Inflammation infiltration	++ Macrophages	++ Macrophages
Liver	Hepatocyte degeneration	+	+
	Inflammation infiltration	+ Lymphocyte	+ Lymphocyte
Kidneys	Necrosis of tubular epithelium	++	++
Multi-organ edema		++	++
Multi-organ congestion		+++	+++

Note: “-”: No lesions (0–5%); “+”: Mild lesions (5–25%); “++”: Moderate lesions (25–50%); “+++”: Severe lesions (≥ 50%)

Discussion

The most significant autopsy finding in insulin overdose fatalities is the presence of subcutaneous or intravenous injection marks. The most common histopathological features include pulmonary edema, congestion, and ischemic-hypoxic changes in neurons [17]. In both cases presented in this study, hemorrhage and inflammation at the injection sites were observed, confirming that these injection injuries were recent rather than old. Neuronal necrosis, as well as congestion, edema, and increased inflammatory cells in multiple organs, were also noted, though these findings are nonspecific. The autopsy alone could not conclusively determine the cause of death. Currently, the most common identification method involves detecting insulin levels using RIA, combined with a comprehensive analysis of the case circumstances to establish the cause of death [17–19]. Nevertheless, insulin degrades rapidly in the body [22], and the results of insulin level measurements can be influenced by various factors, such as hemolysis, making it challenging to accurately detect insulin in postmortem samples. Relying solely on this method to confirm insulin overdose deaths poses significant risks.

Lutz et al. first employed IHC to examine skin from insulin injection sites, detecting insulin-positive staining in the adipocyte membranes and inflammatory cells even after a postmortem interval of up to 24 days. This highlights the importance of immunostaining in identifying insulin overdose cases [23]. Tong et al. detected insulin in peripheral

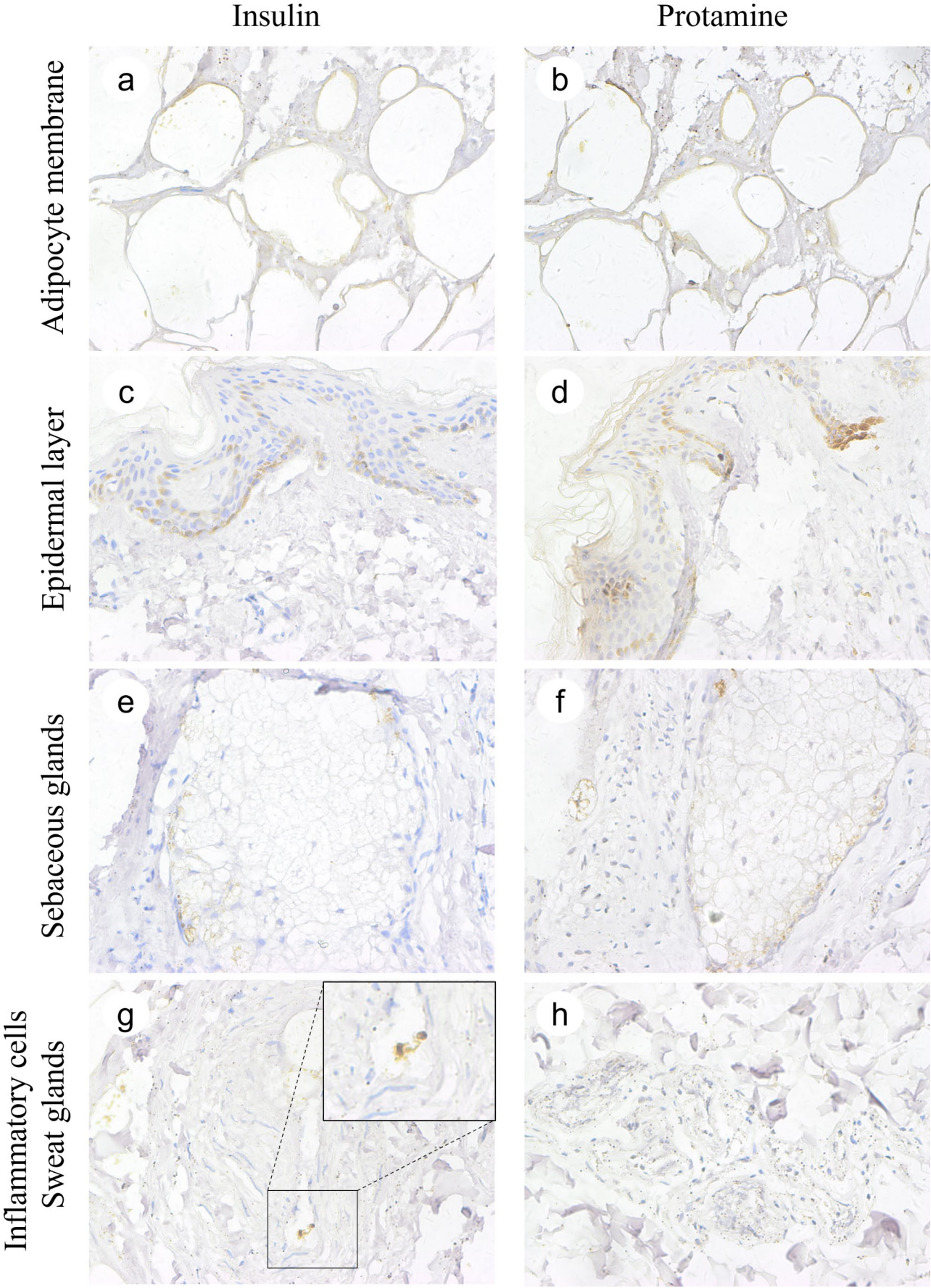
Fig. 5 Distribution of insulin/protamine at injection sites, Insulin/Protamine IHC-stained. **a, b**: Insulin and protamine in the adipocyte membrane of injection sites, 1:400; **c, d**: Insulin and protamine in the epidermal layer and cells, 1:400; **e, f**: Insulin and protamine in sebaceous glands, 1:400; **g**: Insulin in inflammatory cells, 1:400, 1:1000; **h**: Protamine in subcutaneous tissues and sweat glands, 1:400

ganglia and observed protamine in areas where insulin was distributed [24]. Similarly, Ikeda et al. identified insulin in the skin at injection sites, as well as in neuronal cells, renal tubular cells, pituitary gland, and pancreas [25].

In this study, we used IHC/IF to determine the distribution of insulin in injection site skin. Consistent with previous research, insulin and protamine were observed in adipocyte membranes, interstitial spaces, and inflammatory cell cytoplasm. We also observed deposits in the epidermal layer, within epidermal cells, sebaceous glands, and sweat glands. In sebaceous glands, insulin predominantly accumulated at the periphery, with minimal central deposition. Insulin and protamine appeared predominantly in irregular patchy patterns in these areas, though granular distributions were also noted in connective tissue and sweat glands. The almost complete overlap of insulin and protamine distribution confirms that insulin preparations containing protamine had been administered to both deceased individuals. As protamine is an entirely exogenous substance, its presence indirectly confirms the administration of insulin preparations, though this depends on the formulation used. In conclusion, IHC/IF results at the injection site provide direct evidence of exogenous insulin injection, making it a practical method for forensic identification [26–28].

We also used RIA to quantify insulin and C-peptide in the samples. Under normal conditions, the ratio of endogenous insulin to C-peptide is 1:1 [3, 29]. Exogenous insulin reduces C-peptide levels via negative feedback, making the insulin-to-C-peptide ratio a valuable indicator for identifying insulin overdose before hemolysis [9]. Marks et al. reported plasma insulin concentrations in 41 overdose cases, ranging from 0.234 to 323.6 $\mu\text{mol/L}$ [9]. Iwase and Kobayashi studied 18 control cadaveric cases and found average insulin values of $8.1 \pm 5.1 \mu\text{U/mL}$ ($0.058 \pm 0.037 \text{ nmol/L}$) and C-peptide concentrations between 55 and 390 pmol/L ($0.2\text{--}1.4 \text{ ng/mL}$) [30].

In Case 1 of this study, insulin was still detectable in blood 28 days postmortem at $15.6 \mu\text{U/mL}$. Insulin was also detected in urine and vitreous humor in both cases. In Case 2, bile insulin levels were remarkably high at $300 \mu\text{U/mL}$, likely due to the liver's role in insulin clearance. Unfortunately, cerebrospinal fluid was not sampled. Insulin concentrations were elevated in the skin at the injection sites and subcutaneous fat, with higher levels in the outer fat layers due to proximity to the injection site. The insulin-to-C-peptide ratios exceeded 1, suggesting exogenous insulin



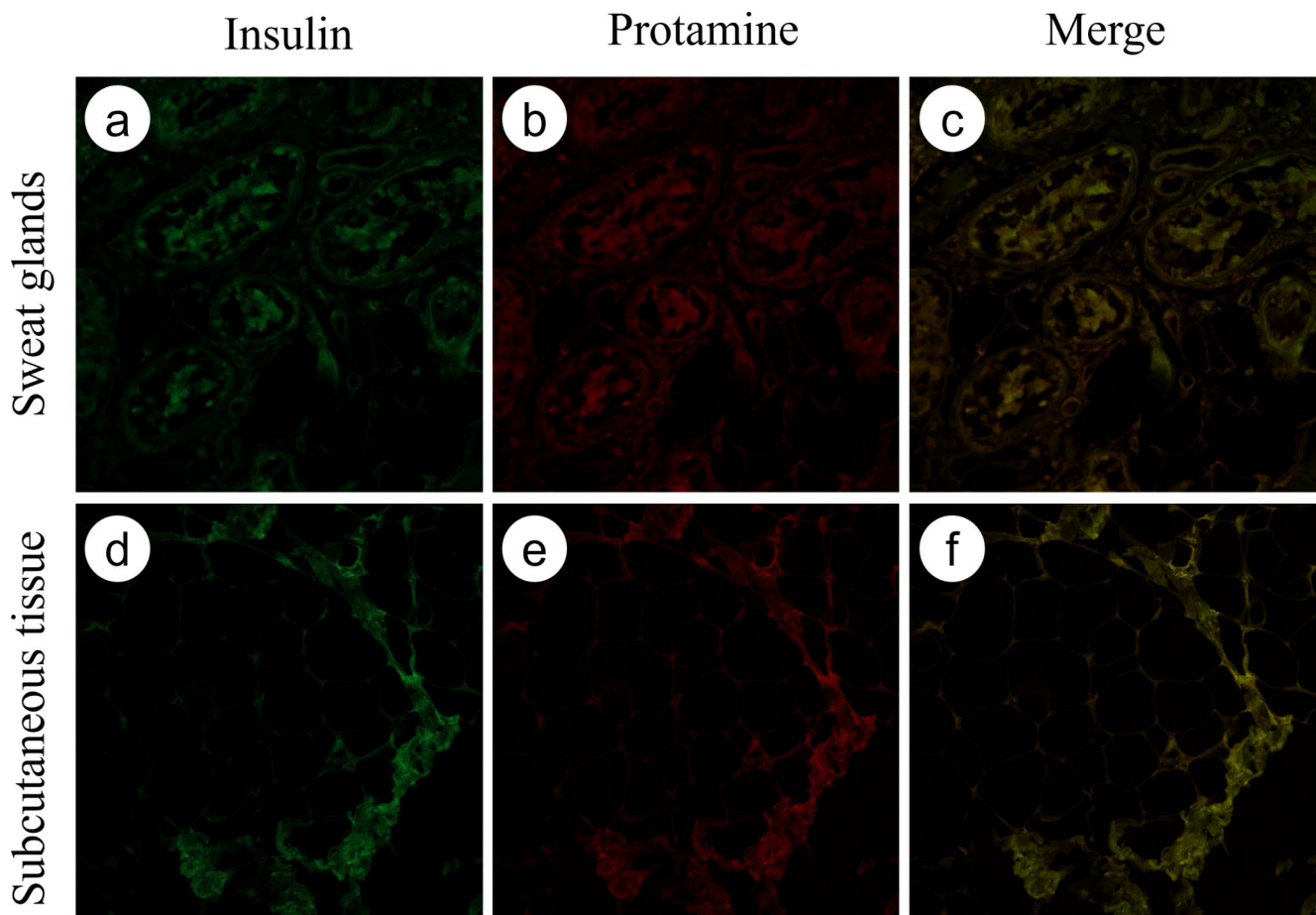


Fig. 6 Insulin and protamine IF-stained. **a, b, c:** Insulin and protamine in sweat glands, 1:200 (**a:** Insulin; **b:** Protamine; **c:** Merge); **d, e, f:** Insulin and protamine in subcutaneous tissue, 1:100 (**d:** Insulin; **e:** Protamine; **f:** Merge)

intake. These RIA results align with IHC/IF findings, providing comprehensive evidence for the causes of death in both cases.

Other drugs, particularly sedatives and hypnotics, should also be considered. Stephenson et al. reviewed 40 insulin overdose cases, attributing 82.5% to insulin toxicity alone, with 17.5% involving mixed drug toxicity [18]. In both cases of this study, psychotropic drugs were detected in the blood of deceased persons. Such cases often involve individuals with histories of depression and long-term medication use, or in cases of insulin homicide, sedatives may be administered by perpetrators to facilitate the crime.

In summary, for cases suspected to involve insulin overdose as the cause of death, we propose the following identification recommendations:

1. Conduct a thorough search of the crime scene, paying attention to the presence of insulin vials, insulin pens, syringes, and other related items, followed by subsequent toxicological testing.
2. Perform the autopsy as soon as possible, as decomposition can increase the difficulty of detecting insulin. Particular attention should be given to the first autopsy. For suspected insulin injection sites, carefully layer the skin and subcutaneous fat to check for signs of subcutaneous hemorrhage and inflammation. It is essential to distinguish between chronic injection injuries (especially in diabetic patients' abdomens) and fresh lethal injection injuries during the autopsy.
3. Collect skin and subcutaneous tissue samples from the injection sites for experimental analysis, and take normal skin from distant areas as control samples. Use IHC/IF to examine the distribution of insulin and protamine in these samples.
4. Collect various body fluid samples (including blood, urine, vitreous humor, cerebrospinal fluid, and bile) and store them frozen. Measure the glucose, insulin, and C-peptide concentrations in all samples.
5. Rule out other causes of death. Consider the possibility of death due to combined insulin and drug toxicity.

Table 3 Results of postmortem toxicology

Sample		Case 1		Case 2	
		Concentration	The post-mortem interval of the sampling	Concentration	The post-mortem interval of the sampling
Insulin in tissue ($\mu\text{U/g}$)	Injection skin	>388.5	About 28d(<1d at room temperature and 28d in frozen)	>3000	9d (about 2d at room temperature and 7d in frozen)
	Subcutaneous adipose	227.95(inner layer)		/	
		>3000(mid-layer)		/	
		>3000(outer layer)		/	
Insulin in humor ($\mu\text{U/mL}$) ^a	Control skin	64		10	
	Blood	15.6		/	
	Urine	118		33.9	
	Vitreous humor	/		6.3	
C-peptide in humor (ng/mL) ^b	Bile	/		>300	
	Blood	0.08		/	
	Urine	0.04		3.54	
	Vitreous humor	/		<0.01	
Others	Bile	/		<0.01	
	blood	7-Aminoclonazepam: (+)		Estazolam: 0.081	
				Ethanol: 5.86	
					About 2d at room temperature

Note: ^a: Reference values: 1.8–11.8 $\mu\text{U/mL}$; ^b: Reference values: 0.78–5.19ng/mL; /: Not detected

Conclusion

We report two cases of insulin-induced homicide. This study used IHC/IF to demonstrate insulin deposition in various locations at the injection sites, including the epidermal layer, epidermal cells, adipocyte membranes, interstitial tissues, inflammatory cells, and glands. The distribution of insulin closely matched that of protamine, providing strong evidence for the administration of exogenous insulin. Significant insulin levels were also detected in the skin, subcutaneous fat, blood, urine, vitreous humor, and bile. This study advances forensic methods for identifying insulin overdose and offers recommendations to support further research in insulin detection.

Key points

1. Two cases of fatal insulin homicide were reported.
2. Histopathological features of fatal insulin overdose were summarized.
3. The distribution of insulin and protamine in the injection site skin was determined through immunohistochemistry and immunofluorescence double-labeling techniques.

4. Quantify insulin concentrations in the injection site and various body fluid samples.
5. Propose identification recommendations for cases of insulin overdose.

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Declarations

Compliance with ethical standards The case was approved by the ethics committee of Huazhong University of Science and Technology. Written informed consent was obtained from the family members of the patient.

Conflict of interest The authors declare that they have no conflict of interest.

Artificial intelligence statement We did not use any artificial intelligence tools to assist in writing.

Informed consent Written informed consent was obtained from the family members of the patient.

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